Supplementary Material for "Fubini–Study Length Bounds on Particle Creation in Driven Bosonic Systems" (A. M. Tishin): Computational Details

S1. Bogoliubov–Valatin equations

The parametric oscillator $\hat{H}(t) = p^2/2 + \Omega^2(t)x^2/2$ has Bogoliubov coefficients u(t), v(t) relative to the instantaneous vacuum satisfying Eq. (11), with initial condition u(0) = 1, v(0) = 0. Independently, we integrate the classical mode equation $\ddot{f} + \Omega^2(t)f = 0$ with $f(0) = 1/\sqrt{2\Omega_0}$, $\dot{f}(0) = -i\Omega_0 f(0)$, and evaluate $N_{ex} = (|\dot{f}|^2 + \Omega^2|f|^2)/(2\Omega) - 1/2$. The two integrations agree to the comparison accuracy 10^{-8}, set by the output interpolation; the Wronskian $\mathrm{Im}(f^*\dot{f})$ is conserved to 6×10^{-11} and $|u|^2 - |v|^2 = 1$ to 2×10^{-11}. Integration: scipy.integrate.solve_ivp, method RK45, rtol = 10^{-11}, atol = 10^{-13}.

S2. Validity tests (Fig. 1)

All tests integrate the mode equation $\ddot{f} + \Omega^2(t)f = 0$ from the instantaneous vacuum with the settings of S1 (rtol = 10^{-11}, atol = 10^{-13}). (a) Modulation $\Omega = 1 + 0.02 \sin \omega_m t$ over t ∈ [0, 100π]; at ω_m = 2 the numerical $N_{ex}(100\pi)$ = 133.25 agrees with the Mathieu estimate $\sinh^2(0.01 \cdot 100\pi)$ = 133.37. (b) tanh ramp from 1 to 0.5 centred at t_0 = 3 with τ = 0.3, then constant frequency up to t = 20. (c) tanh ramps from 1 to r ∈ {0.5, 0.25, 2} with τ ∈ {0.05, 0.1, 0.2, 0.3, 0.5, 0.8, 1.2, 2} (and τ = 5 for Corollary 1), centred at $t_0 = 6\tau + 2$ and integrated to $2t_0 + 5$; the sudden-quench reference is $N_{sudden} = (1 - r)^2/(4r)$.

S3. Multimode and field tests

The mode matrix F(t), defined by $\hat{x}(t) = F\hat{a}_0 + \bar{F}\hat{a}_0^\dagger$, is integrated from $F(0) = A(0)^{-1/2}/\sqrt{2}$, $\dot{F}(0) = -iA(0)F(0)$ with scipy.integrate.solve_ivp (RK45, rtol = 10^{-10} for the multimode tests, atol = 10^{-12}). The Bogoliubov matrix relative to the instantaneous ground state is $\beta = (A^{1/2}\bar{F} + iA^{-1/2}\dot{\bar{F}})/\sqrt{2}$, $N_{tot} = \|\beta\|_F^2$, and the relative squeezing parameters are r_k = arsinh of the singular values of β, from which d_G follows. B(t) is evaluated from central differences of $A(t) = V(t)^{1/2}$ with step 10^{-5}, and the integrals defining Z, L_{FS} and Θ use the trapezoidal rule on 3000 time points. The rotation matrix is R(φ) = [[cos φ, −sin φ], [sin φ, cos φ]]. For the chain V = c(t)I + K (K tridiagonal with diagonal 0.4 and off-diagonal −0.2, c(t) = [1 + 0.25(1 + tanh((t − 3)/0.3))]²,

integrated to t = 8) the normal modes of K are fixed, and each is integrated separately; for M = 50, 200 and 1000 the excitation is 0.82, 3.29 and 16.5, the sum of the single-mode bound (14) over the normal modes is 1.377, 5.51 and 27.6, and the squared right side of Eq. (17) is 1.384, 5.54 and 27.7, while Eq. (16) gives 2.1, 26 and 8.7×10^3. For Eq. (27) the mode equation is integrated from conformal time $-(40/\rho + 20)$ to $40/\rho + 20$ and agrees with Eq. (27) to better than 10^{-8} relative for $\rho = 2$ and 10; for $\rho = 0.5$ and $k \geq 1$, where the particle number falls below 10^{-7}, the comparison is limited by double precision. For Theorem 3, W, α, β and Z are integrated together (DOP853, rtol = 10^{-9}, atol = 10^{-11} for two modes; rtol = 10^{-11} for one mode), and Φ is the running maximum of $\|Z\|$ on 3000 (two modes) or 2×10^5 (one mode) time points. The off-resonant two-mode test of Theorem 3 uses $V_{11} = 1, V_{22} = 2.25$ and J = 0.08cos(0.9t) up to t = 80. For the field modes of Section 3.4, each mode is integrated over conformal time $-(30/\rho + 5)$ to $30/\rho + 5$ with Φ sampled at 20001 points, and the density integrals use 21 points in $0 \leq k \leq 5$ and 40 logarithmically spaced points in $5.3 \leq k \leq 300$. For the moving defect the mode matrix of the ring is integrated with DOP853 (rtol = 10^{-9}, atol = 10^{-11}), B(t) is computed from the time derivative of V in the instantaneous eigenbasis on a grid of 100 points per unit time, and W, Z, α and β for Theorem 3 are integrated with a fourth-order Runge–Kutta step of 0.004; the two integrations give the same final excitation to four digits. The defect starts at $x_a = M/4$, and the motion is followed by 2 time units at rest. The sum of single-mode bounds is $\Sigma_k \sinh^2(\frac{1}{2}\int |\dot{\omega}_k/\omega_k| dt)$ over the instantaneous normal frequencies ω_k ordered by size. Two-mode protocols of Section 3.2: the rotations use $V(t) = R(\varphi(t))\mathrm{diag}(1, 9)R(\varphi(t))^T$, with normal-mode frequencies 1 and 3 and $\varphi(t) = \kappa\tanh(t - 3)$ for $\kappa = 0.5$, 2 and 10, integrated to t = 8; the other protocols have $V_{11} = 1$ and $V_{22} = 2.25$ (frequencies 1 and 1.5) with, respectively, $V_{11} = 1 + 0.05\sin(2t)$ and J = 0.3 up to t = 60, $J(t) = 0.45[1 + \tanh((t - 3)/0.05)]$ up to t = 8, and $J = 0.08\cos(2.5t)$ up to t = 80. Random tests of Theorem 1: 95 protocols for one to three modes, $h(t) = h_0 + a(t)P$ with $h_0 = QQ^T + c\cdot1$ for a Gaussian random matrix Q (c = 1.5 or 2), random symmetric P including x–p blocks, periodic a(t) with frequencies 0.3–3 or tanh steps of width 0.005–1, h(t) kept positive definite; both inequalities of Eq. (15) hold to integration accuracy, and the ratio of the excitation to the length bound reaches 0.9996 for the fastest quenches. Random tests of Theorem 2 in the frame $S_t = (2\gamma_t)^{1/2}$: 12 protocols for two and three modes with x–p blocks, $h(t) = h_0 + a(t)P$ with periodic or tanh a(t); the symplectic frame is computed from $2\gamma_t = h^{-1/2}|h^{1/2}Jh^{1/2}|h^{-1/2}$, B is the part of $-S_t^{-1}\dot{S}_t$ that anticommutes with J, and the Heisenberg propagator is integrated with rtol = 10^{-10}. In all cases $\|B\|_F = \sqrt{2}v_{FS}$ to 10^{-11}, and the excitation never exceeds Eqs. (16) and (17) by

more than 2×10^{-4} of the bound, the same excess as over the proved Eq. (15) and caused by the time discretization near t = 0. Random tests of Theorem 3: 40 protocols for two and three modes, $V(t) = V_0 + a(t)P$ with $V_0 = QQ^T + 1$, $\|P\|_{op} = 1$, amplitude 0.05–0.6 times the smallest eigenvalue of V_0, periodic modulation at frequencies 0.2–4 or a tanh step of width 0.03–3, integrated to t = 12. The random protocols use numpy.random.default_rng with seeds 7 and 11 (Theorem 1), 3 (Theorem 2) and 5 (Theorem 3); the scripts are in the code archive. For Corollary 2, X is computed in the eigenbasis of A and its time derivative by central differences with step 10^{-5}.